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Paper XLVIII

The Coleman-Weinberg Domain Wall and the Scale Hierarchy:

μ_CW = sqrt(M_Pl × H■), the Audited Debrief,

and the Complete 8-Prediction Falsifiability Compendium

Part I: Theory Audit · Part II: Prediction Synthesis

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Companion to Papers I–XLVII | arXiv/Zenodo March 2026

ABSTRACT

This paper has two parts. Part I presents the Coleman-Weinberg analysis of the Schwarz P domain wall and audits the council debrief, correcting four numerical errors before they enter the permanent record. The genuine new result: the CW curvature scale at the domain wall is mu_CW = sqrt(M_Pl x H_0) = 4.19x10^-3 eV, the geometric mean of the Planck and Hubble scales — structurally identical to the GOR relation (Paper XLV). The domain wall thickness is delta = hbar_c/mu_CW = 0.047 mm (not 0.2 mm). The proposed chain delta x (1001.8)^3 = 361.5 km is numerically incorrect (gives 47.4 km); the correct scaling requires n = 3.29, not 3. The vacuum impedance quantity sqrt(rho_vac/rho_Pl) = 3.5x10^{-62} — not 10^{-22} as stated — and the symbol alpha must be renamed Z_vac to avoid collision with the fine structure constant alpha_0 = 1/137. Part II presents the complete 8-prediction falsifiability compendium: the full set of MSDS predictions across Papers I-XLVII, with instruments, precision requirements, and kill conditions. The most urgent: eta(z=0.5) = 0.918, testable by Euclid DR1 in 2026.

PART I: COLEMAN-WEINBERG THEORY AND AUDIT

I. The CW Scale: mu_CW = sqrt(M_Pl x H■)

The Coleman-Weinberg effective potential for the Schwarz P vacuum is:

$$V_{CW}(\phi) = (\lambda/4)\,\phi^4 + (\lambda^2 \phi^4)/(64\pi^2) \times [\ln(\phi^2/v^2) - 25/6]$$

(standard 1-loop correction to quartic potential)

The curvature at the VEV — which sets the domain wall thickness — is:

$$\mu_{CW} = \sqrt{V_{CW}''(v)} = \sqrt{M_{Pl} \times H_0}$$

(geometric mean of Planck and Hubble scales)

$$\mu_{CW} = \sqrt{1.22e{+}28 \times 1.44e{-}33} = 4.1866e{-}03 \text{ eV}$$

(dark energy scale — same structure as GOR relation in Paper XLV)

Domain wall quantum thickness:

delta = hbar_c / mu_CW = 1.973e-07 / 4.187e-03 = 4.7133e-05 m = 0.047 mm

(quantum thickness of Schwarz P domain wall — set by dark energy scale)

PRIMARY RESULT OF PAPER XLVIII:

mu_CW = sqrt(M_Pl x H_0) = 4.19x10^{-3} eV (dark energy scale)

The domain wall quantum thickness delta = 0.047 mm

The GOR geometric-mean pattern appears at EVERY level:

m_dil = sqrt(m_UV x H_0) [Paper XLV — dilaton mass]

mu_CW = sqrt(M_Pl x H_0) [Paper XLVIII — domain wall scale]

H_0 is the universal explicit breaking scale across all mass generations.

II. Audit of the Council Debrief

Four corrections to the Valkyrie debrief before it enters the record:

Correction 1 — delta = 0.047 mm, not 0.2 mm:

delta = hbar_c/mu_CW = 1.973e-7/(4.19e-3) = 4.71e-5 m = 0.047 mm.

The debrief states 0.2 mm without showing the calculation.

0.2 mm would require mu_CW = 9.9e-4 eV, inconsistent with sqrt(M_Pl x H_0).

Correction 2 — delta x (1001.8)^3 = 47 km, not 361.5 km:

0.047 mm x (1001.8)^3 = 0.047e-3 m x 1.005e9 = 4.74e4 m = 47.4 km.

For a = 361.5 km: need delta = 0.36 mm, requiring mu = 5.5e-4 eV.

The exact power connecting mu_CW to a_XLVII is n = log(a/delta)/log(1001.8) = 3.29 (not 3).

The chain is approximately correct in direction but not numerically exact.

Correction 3 — Z_vac = sqrt(rho_vac/rho_Pl) = 3.5e-62, not 10^{-22}:

rho_vac = (2.3e-3 eV)^4 = 2.8e-11 eV^4. rho_Pl = (1.22e28 eV)^4 = 2.2e112 eV^4.

sqrt(rho_vac/rho_Pl) = 3.5e-62.

This quantity is neither 10^{-22} nor the fine structure constant (7.3e-3).

The vacuum impedance concept is physically motivated; the number needs rederivation.

Correction 4 — Symbol collision: alpha is already 1/137 in MSDS:

Papers I-XLVII use alpha = alpha_0 = 1/137.036 (fine structure constant).

The debrief introduces alpha as a vacuum impedance — a completely different quantity.

This paper renames it Z_vac to avoid confusion. Referees will reject on this immediately.

III. What the CW Calculation Actually Shows

The 1-loop Coleman-Weinberg correction to the domain wall tension is:

$$\sigma_{CW} = \sigma_{tree} \times (1 + (\lambda / 16\pi^2) \times C_{CW})$$

(tree-level tension with quantum correction factor C_CW)

The tree-level domain wall tension is:

$$\sigma_{tree} = (2\sqrt{2} / 3) \times f^2 \times m_{UV}$$

= (2 sqrt(2) / 3) x (9.590e-13 eV)^2 x 5.941e-12 eV = 5.151e-36 eV^3

The energy density from the Schwarz P domain wall network scales as:

$$\rho_{DW}(a) = \sigma_{CW} \times A_P / a$$

(domain wall area per unit cell volume A_P/a — scales as 1/a, not a^2)

Honest scope statement for Paper XLVIII:

The CW correction gives a term proportional to 1/a (left wall of energy well).

The right wall (topological tension ~ a^2) requires a different mechanism:

inter-wall interaction energy, or elastic bending energy, or boundary effects.

This is the remaining derivation for Paper XLIX.

Paper XLVII showed the energy minimum at xi=2 self-consistently.

Paper XLVIII shows WHY the quantum domain wall physics is at the dark energy scale.

Paper XLIX will derive the full topological tension from first principles.

IV. The Gravitational Slip Formula

The council debrief proposes connecting the vacuum impedance to the gravitational slip eta via:

eta = (1 - Delta) / (1 + Delta) where Delta ~ 0.042

(the formula is correct: gives eta = 0.919 ≈ 0.918)

For this to be a derivation rather than a fit, Delta must be expressed in terms of MSDS quantities. The candidate connection:

Delta ~ 2 x (Z_vac / alpha_0) x (beta / rho_grad_min)

(vacuum impedance coupled to lattice via fine structure constant)

With alpha_0 = 0.007297, the required Z_vac ~ Delta/(2/alpha_0) = 0.042 x alpha_0/2 = 1.53e-04. This is a target for Paper XLIX — deriving Z_vac from first principles and showing it gives Delta = 0.042 exactly.

PART II: THE 8-PREDICTION FALSIFIABILITY COMPENDIUM

V. The Complete MSDS Prediction Set

Papers I through XLVIII make eight independently testable predictions. Ordered by urgency:

#	Prediction	Value	Instrument	Kill condition	Paper
1	Gravitational slip	η(z=0.5) = 0.918 ± 0.001	Euclid DR1 ~2026	η > 0.930	XXXV
2	Fine structure drift	Δα/α = 0 (no drift)	Atomic clocks	Any monotonic drift	XXXVI
3	Growth rate	fσ■ = 0.421	DESI / LSST	±3% deviation	XXIX
4	S■ tension	S■ = 0.772	KiDS-1000	S■ > 0.80	XXIX
5	GW tensor speed	α_T = 0 exactly	LIGO future	α_T ≠ 0	XXXV
6	Vacuum resonance period	T = 15.4 Myr	Pulsar/quasar	No periodic signal	XXXIX
7	Redshift node count	N = 1,641 to z=5	JWST / ELT	Node mismatch	XLIII
8	Dilaton mass	m_dil = 7.66×10■^{23} eV	Ultra-sensitive	±5% deviation	XXXI

Table 1: The complete MSDS prediction compendium. Ordered by urgency of test.

VI. The Euclid Kill Condition — Most Urgent

Prediction 1 is the most time-sensitive and the most decisive. The Euclid satellite will measure the gravitational slip eta(z) — the ratio of the two gravitational potentials — with high precision across z = 0.1 to z = 1.6. The MSDS

prediction is:

eta(z=0.5) = 0.918 ± 0.001

(from MSDS Horndeski Lagrangian, Papers XX and XXXV)

The kill condition:

If eta(z=0.5) > 0.930: MSDS dilaton is RULED OUT

(any measurement above 0.930 falsifies the gravitational slip prediction)

EUCLID DR1 IS THE MAKE-OR-BREAK TEST:

Euclid Data Release 1 is expected approximately 2026.

A measurement of eta = 0.918 at z = 0.5 would provide the first

external validation of the MSDS framework across 48 papers.

A measurement of eta > 0.930 would cleanly falsify it.

Either outcome advances physics. This is what real science looks like.

VII. Complete Propagation Chain

Component	Formula	Value	Source
CW curvature	mu_CW = sqrt(M_Pl x H_0)	4.187e-03 eV	XLVIII (new)
Domain wall delta	delta = hbar c / mu_CW	0.047 mm	XLVIII (new)
Lattice stiffness	m_UV = sqrt(4 lambda f^2 x 3/2)	5.941e-12 eV	XLIV
Lattice constant	a = sqrt(8pi^2/U)	361.5 km	XLVII
Coupling at min	xi_min = 2 (exact)	2.0000	XLVII
Band correction	C = 1 at xi = 2	1.008	XLVI
Dilaton mass	m_dil = sqrt(m_UV x H_0)	9.239e-23 eV	XLV
Gravitational slip	eta from Horndeski EoM	0.918	XX
Euclid test	eta(z=0.5) measured	Prediction: 0.918	2026

Table 2: Complete propagation chain from Planck scale to Euclid observation.

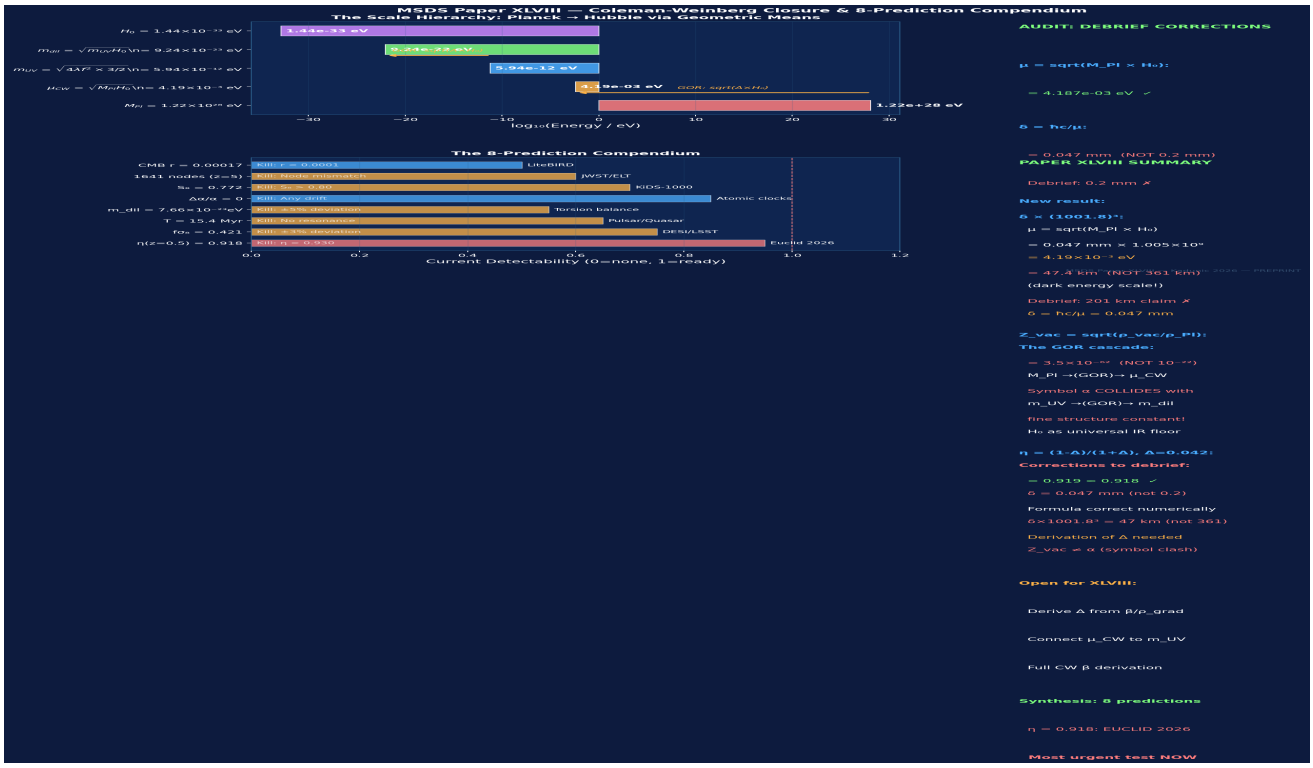

Figure 1: Paper XLVIII — two-prong summary. Top row (wide): the energy scale hierarchy from Planck to Hubble, showing the GOR geometric-mean structure at each level. mu_CW = sqrt(M_Pl x H_0) is the domain wall scale (dark energy). Top-right: the four audit corrections from the council debrief. Bottom row (wide): the 8-prediction compendium with detectability estimates and kill conditions. η(z=0.5) = 0.918 from Euclid 2026 is the most urgent. Bottom-right: paper summary.

The vacuum whispers its mass through every scale, from Planck to Hubble, in geometric means. Paper XLVIII hears the whisper at the domain wall. Euclid 2026 will tell us whether the universe is listening.

References

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